

# MD TANVIR ISLAM

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## EDUCATION AND AWARDS

|                              |                                                                                                                                                                                                                                          |                            |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| September 2025<br>Present    | PhD, KYUNGPOOK NATIONAL UNIVERSITY (KNU), South Korea<br><i>Department of Electronic and Electrical Engineering</i><br>Research Area : Mamba-SSM and Robot Manipulation.                                                                 |                            |
| March 2023<br>June 2025      | MS, SUNGKYUNKWAN UNIVERSITY (SKKU), South Korea<br><i>Department of Computer Science and Engineering</i>   CGPA : 4.19/4.50<br>Thesis : Study Towards Robust Image Enhancement in Adverse Atmospheric Conditions.                        | <a href="#">Transcript</a> |
| March 2022<br>February 2023  | Korean Language and Cultural Program, PAI CHAI UNIVERSITY, South Korea<br><i>Korean Language and Cultural Program</i>   TOPIK Level : 4/6                                                                                                | <a href="#">Transcript</a> |
| January 2015<br>January 2019 | BS, North Western University (NWU), Bangladesh<br><i>Department of Computer Science and Engineering</i>   CGPA : 3.98/4.00<br>Thesis : Study on diabetes prediction for typical and non-typical cases using machine learning approaches. | <a href="#">Transcript</a> |

## RESEARCH EXPERIENCES

IF : IMPACT FACTOR

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| March 2023<br>Present    | Research Assistant, <a href="#">VIS2KNOW LAB</a> , Sungkyunkwan University, South Korea<br><a href="#">Lab Director</a> <ul style="list-style-type: none"><li>➤ Data curation and research in AI image detection, image dehazing, and low-light image enhancement.</li><li>➤ Our research on image dehazing, LLIE, and AI-synthesized image detection has led to three papers are published at top-tier conferences and journals such as <i>ACM Multimedia</i> (<b>Rank A</b>), <i>Asian Conference on Computer Vision (ACCV)</i> (<b>Rank B</b>), <i>Alexandria Engineering Journal (AEJ)</i> (IF : 6.2, Q1). Also, three of our innovations are registered at <i>US Patent Office</i>.</li></ul> |
| July 2018<br>August 2021 | Research Assistant, <a href="#">CSHI-RLAB</a> , North Western University, Bangladesh<br><a href="#">Lab Director</a> <ul style="list-style-type: none"><li>➤ Collected real-life datasets to perform research using machine-learning techniques and <b>published several papers</b> at different international conferences, which achieved around <b>100 citations</b> so far.</li></ul>                                                                                                                                                                                                                                                                                                           |

## TEACHING EXPERIENCES

|                               |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| March 2024<br>August 2024     | Teaching Assistant, DEPARTMENT OF AI, Sungkyunkwan University, South Korea <ul style="list-style-type: none"><li>➤ Assisted with curriculum planning for courses and held office hours to support and mentor students.</li><li>➤ Facilitated group discussions and study sessions to enhance collaborative learning.</li></ul> |
| January 2019<br>December 2020 | Part-time Lecturer, SOUTH-EAST ENGINEERING COLLEGE, Khulna, Bangladesh <ul style="list-style-type: none"><li>➤ Planned, prepared and presented lectures while assessing student assignments.</li><li>➤ Maintained communication with teachers and students.</li></ul>                                                          |

## AWARDS AND SCHOLARSHIPS

|                              |                                                                                                                                                                                                |                             |
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| November 2024                | Academic Excellence Award Winner, NIIED, Government of South Korea<br>The NIIED selected me as one of the three students for this award based on the research outcomes.                        | <a href="#">Certificate</a> |
| December 2023                | 16th Korean Language Society Scholarship, PAI CHAI UNIVERSITY, South Korea<br>Received this award based on my academic performance and on-campus activities including volunteering.            | <a href="#">Certificate</a> |
| August 2022<br>February 2025 | Global Korea Scholarship (GKS), NIID, South Korea<br>Received this award based on my academic performance and on-campus activities including volunteering.                                     | <a href="#">Certificate</a> |
| January 2015<br>January 2019 | First Position in Merit List, NORTH WESTERN UNIVERSITY, Bangladesh<br>Certified by my undergraduate university for securing the 1 <sup>st</sup> position in the merit list among all students. | <a href="#">Certificate</a> |

- SCIE | Q1 | IF : 6.2** **Md Tanvir Islam**, Ik Hyun Lee, Ahmed Ibrahim Alzahrani, and Khan Muhammad\*. "MEXFIC : A meta ensemble eXplainable approach for AI-synthesized fake image classification." *Alexandria Engineering Journal* 116 (2025) : 351-363.
- SCIE | Q1 | IF : 7.8** **Md Tanvir Islam**, Safkat Shahrier Swapnil, Md Masum Billal, Asif Karim\*, Niussha Shafiabady, and Md Mehedi Hassan\*. "Resource constraint crop damage classification using depth channel shuffling." *Engineering Applications of Artificial Intelligence* 144 (2025) : 110117.
- SCIE | Q1 | IF : 6.2** **Md Tanvir Islam**, Abu Saleh Musa Miah, M. Raihan, ... and Khan Muhammad\*. "DiaBD : A novel benchmark dataset for diabetes prediction." *Alexandria Engineering Journal* 116 (2025) : 351-363.
- SCIE | Q1 | IF : 13.7** **Md Tanvir Islam**, Inzamamul Alam, and Simon S. Woo, Khan Muhammad\*. "WeatherIR : Degradation Aware Multi-Weather Image Restoration." *IEEE Transactions on Image Processing (IEEE TIP)*. (In review)
- Rank A** **Md Tanvir Islam**, Inzamamul Alam, Sambit Bakshi, Khan Muhammad, Javier Del Ser, Sangtae Ahn1. "SpikeRain : Towards Energy-Efficient Single Image Deraining with Spiking Neural Networks." *Winter Conference on Applications of Computer Vision, WACV 2026*.
- Rank A** **Md Tanvir Islam**, Nasir Rahim, Saeed Anwar, Muhammad Saqib, Sambit Bakshi, and Khan Muhammad\*. "Hazespace2m : A dataset for haze aware single image dehazing." In Proceedings of the *ACM Multimedia, ACM MM 2024*.
- Rank B** **Md Tanvir Islam**, Inzamamul Alam, Simon S. Woo\*, Saeed Anwar, I. K. Lee, and Khan Muhammad\*. "LoLI-Street : Benchmarking Low-Light Image Enhancement and Beyond." In Proceedings of the *Asian Conference on Computer Vision, ACCV 2024*.
- Rank A** Inzamamul Alam, **Md Tanvir Islam**, and Simon S. Woo\*. "Saliency-Aware Diffusion Reconstruction for Effective Invisible Watermark Removal." *ACM Web Conference, WWW 2025*.
- Rank A** Inzamamul Alam, **Md Tanvir Islam**, Simon S. Woo\*, Khan Muhammad. "SpecGuard : Spectral Projection-based Advanced Invisible Watermarking." *International Conference on Computer Vision, ICCV 2025*.
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- U.S. Patent** **Md Tanvir Islam**, Ik Hyun Lee, Abdul Khader Jilani Saudagar, Abdullah Altameem, Mohammed Abaoud, Khan Muhammad. "Method for image dehazing of vehicular images." **U.S. Patent No. 12277682**, issued April 15, 2025.
- U.S. Patent** **Md Tanvir Islam**, Ik Hyun Lee, Abdul Khader Jilani Saudagar, Abdullah Altameem, Mohammed Abaoud, and Khan Muhammad. "Apparatus and method for single image dehazing considering haze type classification." **U.S. Patent No. 12217398**, issued February 4, 2025.
- U.S. Patent** Abdul Khader Jilani Saudagar, Safkat Shahrier Swapnil, Sandip Kumer Sarker, Arpon Bose Ditya, **Md Tanvir Islam**, Md. Abu-Talha Roni, Hitoun A. Alsagri, and Khan Muhammad. "Adaptive Speed Control For Line-Following Robots and Method Thereof." **U.S. Patent No. 12287650**, issued April 29, 2025.
- U.S. Patent** Shehzad Ali, **Md Tanvir Islam**, Ik Hyun Lee, Abdul Khader Jilani Saudagar, Khan Muhammad. "Method and system for controlling autonomous vehicles." **U.S. Patent No. 12485924**, issued December 2, 2025.
- U.S. Patent** Shehzad Ali, **Md Tanvir Islam**, and Khan Muhammad. "Apparatus and method for child hazardous activity anticipation and risk assessment system." *U.S. Patent (Application Filed)*
- Book Chapter** Rezuana Haque and **Md Tanvir Islam** (2025). "Neural Dynamics : Unravelling the Complexity of Brain Activity." In *Brain Networks in Neuroscience : Personalization Unveiled Via Artificial Intelligence*. Chapter 1, pp. 1-19. (In review)

## SKILLS AND INTERESTS

Language English : [IELTS \(2019\) : 7.0, no band < 6.5.](#)  
Korean : [TOPIK \(2022\) : 4/6.](#)

Programming Python; **Frameworks** : PyTorch, Keras, SpikingJelly; **Database** : PostgreSQL, MySQL.

**Tools** PowerBI, LaTeX, PyCharm, Jupyter, VSCode, Photoshop, Illustrator, Figma, Microsoft Suites.

**Interests** Image Restoration, Invisible Watermarking, Activity Recognition, Deepfake, Bioinformatics.

## PROJECTS

HAZESPACE2M –  [GITHUB.COM/TANVIRNWU/HAZESPACE2M](https://github.com/TANVIRNWU/HAZESPACE2M)

LEAD AUTHOR

A large-scale hazy dataset of over 2 million images across 10 distinct levels and propose a novel framework for haze-aware single image dehazing that significantly improves PSNR by 2.41%, SSIM by 17.14%, and MSE by 10.2% over general dehazers.

[SOTA Image Dehazing Methods](#) [SOTA Classifiers](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Photoshop](#)

TriFUSE & LOLI-STREET –  [GITHUB.COM/TANVIRNWU/TriFUSE](https://github.com/TANVIRNWU/TriFUSE)

LEAD AUTHOR

First to introduce a 33K paired low-light street scene dataset with object detection labels and propose TriFUSE, a transformer-wavelet-diffusion model that achieves SOTA results for real-life low-light image enhancement for autonomous systems.

[LLIE Methods](#) [Diffusion Denoising](#) [Transformers](#) [Deep Learning](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Photoshop](#)

SADRE –  [GITHUB.COM/TANVIRNWU/SADRE](https://github.com/TANVIRNWU/SADRE)

CO-AUTHOR

We introduced SADRE, which is a saliency-aware diffusion reconstruction method for effective invisible watermark removal.

[SOTA Image Watermarking Methods](#) [Attacks](#) [Diffusion](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Illustrator](#)

## TRAININGS AND LEARNING

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| August 2019    | LICT Top-up IT Training, <a href="#">ICD DIVISION</a> , Bangladesh                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">Certificate</a>                                                                                                                                    |
| September 2019 | <i>Bangladesh Computer Council (Certified by George Washington University, Washington, D.C.)</i> <ul style="list-style-type: none"><li>▶ The training includes 240 hours of technical skills such as Java, HTML, CSS, JavaScript, Bootstrap, and MVC Pattern, and 60 hours of other soft skills like presentation and communication.</li></ul>                                                                                                                                                                         |                                                                                                                                                                |
| February 2018  | Google Bangladesh Android StudyJam, <a href="#">GDG DHAKA</a> , Bangladesh                                                                                                                                                                                                                                                                                                                                                                                                                                             | <a href="#">Certificate</a>                                                                                                                                    |
|                | <i>ICT Division, GDG Dhaka, Google Developers</i> <ul style="list-style-type: none"><li>▶ A 225-hour Android app development course with regular mini-project assignments, culminating in a final complete app development project.</li></ul>                                                                                                                                                                                                                                                                          |                                                                                                                                                                |
| January 2019   | Self Learning,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Online, Books↓                                                                                                                                                 |
| Present        | <i>Some of the courses and books I completed to grow my knowledge.</i> <ul style="list-style-type: none"><li>▶ PyTorch for Deep Learning Bootcamp (52 hours)</li><li>▶ Python for Data Science, AI &amp; Development (25 hours)</li><li>▶ Python for Data Science and Machine Learning Bootcamp (25 hours)</li><li>▶ Foundation of User Experience (UX) Design (19 hours)</li><li>▶ Modern Computer Vision with PyTorch</li><li>▶ Python Deep Learning Projects</li><li>▶ Advanced Deep Learning with Python</li></ul> | Udemy   Mar 2024<br>IBM   Mar 2024<br>Udemy   Sept 2019<br>Google   Oct 2021<br>Book by V Kishore Ayyadevara<br>Book by Matthew Lamons<br>Book by Ivan Vasilev |

## OTHER ACTIVITIES

|                                                                                        |                     |
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| Reviewer, <a href="#">IEE Transaction of Image Processing (IEEE TIP)</a>               | September 2025      |
| Reviewer, <a href="#">Information Fusion</a>                                           | August 2025         |
| Reviewer, <a href="#">IEEE Open Journal of the Computer Society</a>                    | May 2025            |
| Reviewer, <a href="#">Engineering Application of Artificial Intelligence</a>           | April 2025          |
| Reviewer, <a href="#">International Conference of Pattern Recognition (ICPR)</a>       | July 2024           |
| Reviewer, <a href="#">PLOS One</a> and <a href="#">PLOS Digital Health</a>             | January 2025        |
| Reviewer, <a href="#">Frontiers in Nutrition</a>                                       | March 2024          |
| Founder of Animal Rescue Team, <a href="#">Khulna Cat Society</a> , Khulna, Bangladesh | July 2016 - Present |

## “ REFERENCES

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**Khan Muhammad | Relation : MS Supervisor**

**Associate Professor**, *Department of Applied Artificial Intelligence*, SUNGKYUNKWAN UNIVERSITY, SOUTH KOREA

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**Saeed Anwar | Relation : Academic Mentor**

**Senior Lecturer**, *Computer Science and Software Engineering*, UNIVERSITY OF WESTERN AUSTRALIA (UWA), AUSTRALIA

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